



Construction Quantity Calculator

Structural Quantity Report

Project: عمارة صادق قبيظة
Client: العميل عبدالله القبيضة
Engineer: المهندس ماجد القبيضة
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Prepared by

Eng. Majid Alqobidhah

Professional Civil Engineer



Concrete Volume

67.99 m³

Steel Weight

599.4 kg

Total Cost

\$7,098.30

Elements

27

Total Concrete

67.99 m³

Total Steel

599.4 kg (0.599 tons)

Grand Total

\$7,098.30

Floor Breakdown

Floor	Concrete (m ³)	Steel (tons)	Cost
الدور الأرضي	10.111	0.101	1,061.11
الدور الأول	20.222	0.202	2,072.22
الدور الثاني	30.333	0.303	3,093.33
TOTAL	67.986	0.599	7,098.30

Isolated Footings

Volume: 7.554 m³ | Steel: 66.6 kg | Cost: \$788.70 | Elements: 3

Label	Dimensions (m)	Qty	Volume (m ³)	Steel (kg)	Cost
أول عمارة Footing	0.11×0.22×1.33	1	1.111	11.1	116.65
Footing R2	0.11×0.23×1.43	2	2.444	22.2	255.50
Footing R3	0.11×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70

Strip Footings

Volume: 7.554 m³ | Steel: 66.6 kg | Cost: \$788.70 | Elements: 3

Label	Dimensions (m)	Qty	Volume (m ³)	Steel (kg)	Cost
أول Strip عمارة	0.12×0.22×1.33	1	1.111	11.1	116.65
Strip R2	0.12×0.23×1.43	2	2.444	22.2	255.50
Strip R3	0.12×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70

Raft Foundation

Volume: 7.554 m³ | Steel: 66.6 kg | Cost: \$788.70 | Elements: 3

Label	Dimensions (m)	Qty	Volume (m ³)	Steel (kg)	Cost
أول Raft عمارة	0.13×0.22×1.33	1	1.111	11.1	116.65
Raft R2	0.13×0.23×1.43	2	2.444	22.2	255.50
Raft R3	0.13×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70



Columns				Volume: 7.554 m³ Steel: 66.6 kg Cost: \$788.70 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
أول عمارة Column R2	0.14×0.22×1.33	1	1.111	11.1	116.65
Column R2	0.14×0.23×1.43	2	2.444	22.2	255.50
Column R3	0.14×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70

Beams				Volume: 7.554 m³ Steel: 66.6 kg Cost: \$788.70 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
أول عمارة Beam R2	0.15×0.22×1.33	1	1.111	11.1	116.65
Beam R2	0.15×0.23×1.43	2	2.444	22.2	255.50
Beam R3	0.15×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70

Solid Slabs				Volume: 7.554 m³ Steel: 66.6 kg Cost: \$788.70 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
أول عمارة SolidSlab R2	0.16×0.22×1.33	1	1.111	11.1	116.65
SolidSlab R2	0.16×0.23×1.43	2	2.444	22.2	255.50
SolidSlab R3	0.16×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70

Hollow Block Slabs				Volume: 7.554 m³ Steel: 66.6 kg Cost: \$788.70 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
أول عمارة HollowSlab R2	0.17×0.22×1.33	1	1.111	11.1	116.65
HollowSlab R2	0.17×0.23×1.43	2	2.444	22.2	255.50
HollowSlab R3	0.17×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70

Flat Slabs				Volume: 7.554 m³ Steel: 66.6 kg Cost: \$788.70 Elements: 3	
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
أول عمارة FlatSlab R2	0.18×0.22×1.33	1	1.111	11.1	116.65
FlatSlab R2	0.18×0.23×1.43	2	2.444	22.2	255.50
FlatSlab R3	0.18×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70



Waffle Slabs			Volume: 7.554 m³ Steel: 66.6 kg Cost: \$788.70 Elements: 3		
Label	Dimensions (m)	Qty	Volume (m³)	Steel (kg)	Cost
أول WaffleSlab عمارة	0.19×0.22×1.33	1	1.111	11.1	116.65
WaffleSlab R2	0.19×0.23×1.43	2	2.444	22.2	255.50
WaffleSlab R3	0.19×0.24×1.53	3	3.999	33.3	416.55
TOTAL			7.554	66.6	788.70

Bill of Quantities (BOQ)

عمارة صادق قبيظة

#	Type	Floor	Label	L×W×H (m)	Qty	Vol/unit	Total Vol	Steel kg	Cost
1	F	الدور 1	عمارة Footin	0.11×0.22×1.33	1	1.111	1.111	11.1	116.65
2	F	الدور 1	Footing R2	0.11×0.23×1.43	2	1.222	2.444	22.2	255.50
3	F	الدور 1	Footing R3	0.11×0.24×1.53	3	1.333	3.999	33.3	416.55
4	SF	الدور 1	عمارة Strip	0.12×0.22×1.33	1	1.111	1.111	11.1	116.65
5	SF	الدور 1	Strip R2	0.12×0.23×1.43	2	1.222	2.444	22.2	255.50
6	SF	الدور 1	Strip R3	0.12×0.24×1.53	3	1.333	3.999	33.3	416.55
7	RF	الدور 1	عمارة Raft أ	0.13×0.22×1.33	1	1.111	1.111	11.1	116.65
8	RF	الدور 1	Raft R2	0.13×0.23×1.43	2	1.222	2.444	22.2	255.50
9	RF	الدور 1	Raft R3	0.13×0.24×1.53	3	1.333	3.999	33.3	416.55
10	C	الدور 2	عمارة Column	0.14×0.22×1.33	1	1.111	1.111	11.1	116.65
11	C	الدور 2	Column R2	0.14×0.23×1.43	2	1.222	2.444	22.2	255.50
12	C	الدور 2	Column R3	0.14×0.24×1.53	3	1.333	3.999	33.3	416.55
13	B	الدور 2	عمارة Beam أ	0.15×0.22×1.33	1	1.111	1.111	11.1	116.65
14	B	الدور 2	Beam R2	0.15×0.23×1.43	2	1.222	2.444	22.2	255.50
15	B	الدور 2	Beam R3	0.15×0.24×1.53	3	1.333	3.999	33.3	416.55
16	SS	الدور 2	عمارة SolidS	0.16×0.22×1.33	1	1.111	1.111	11.1	116.65
17	SS	الدور 2	SolidSlab R2	0.16×0.23×1.43	2	1.222	2.444	22.2	255.50
18	SS	الدور 2	SolidSlab R3	0.16×0.24×1.53	3	1.333	3.999	33.3	416.55
19	HS	الدور 3	عمارة Hollow	0.17×0.22×1.33	1	1.111	1.111	11.1	116.65
20	HS	الدور 3	HollowSlab R	0.17×0.23×1.43	2	1.222	2.444	22.2	255.50
21	HS	الدور 3	HollowSlab R	0.17×0.24×1.53	3	1.333	3.999	33.3	416.55
22	FS	الدور 3	عمارة FlatSl	0.18×0.22×1.33	1	1.111	1.111	11.1	116.65
23	FS	الدور 3	FlatSlab R2	0.18×0.23×1.43	2	1.222	2.444	22.2	255.50
24	FS	الدور 3	FlatSlab R3	0.18×0.24×1.53	3	1.333	3.999	33.3	416.55
25	WS	الدور 3	عمارة Waffle	0.19×0.22×1.33	1	1.111	1.111	11.1	116.65
26	WS	الدور 3	WaffleSlab R	0.19×0.23×1.43	2	1.222	2.444	22.2	255.50
27	WS	الدور 3	WaffleSlab R	0.19×0.24×1.53	3	1.333	3.999	33.3	416.55
TOTAL						67.986	599.4	7,098.30	

Concrete Mix Design — Material Quantities

Mix Ratio 1 : 2 : 4 (Cement : Sand : Gravel) • Compaction factor 1.54

Total Concrete Volume

67.986 m³

Dry volume required (×1.54): 104.698 m³

Material Quantities

Material	Volume (m³)	Equivalent Weight	Ratio Part
Cement	14.957	431 bags × 50 kg	1
Sand	29.914	47,862 kg	2
Gravel	59.828	92,733 kg	4
TOTAL DRY VOLUME	104.698	104.698	1 : 2 : 4

Per 1 m³ of Concrete — Reference Quantities

Material	Quantity	Details
Cement	7 bags	0.2200 m³ • 50 kg/bag
Sand	0.4400 m³	704 kg
Gravel	0.8800 m³	1,364 kg

Total Concrete

Total Steel

Grand Total Cost

67.986 m³

599.4 kg (0.599 tons)

\$7,098.30